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CONDUCTION MODE CONTROL

Abstract

A circuit includes an input terminal, a reference voltage circuit, a comparator, a delay circuit, a conduction mode control circuit, and a switching converter controller. The input terminal is configured to receive a converter output voltage. The reference voltage circuit has an output. The comparator has a first input coupled to the input terminal, a second input coupled to the output of the reference voltage circuit, and an overvoltage output. The delay circuit has an input coupled to the overvoltage output, and a delayed overvoltage output. The conduction mode control circuit has a first input coupled to the overvoltage output, a second input coupled to the delayed overvoltage output, and a mode output. The switching converter controller has a continuous conduction mode input coupled to the mode output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/559,496, filed Feb. 29, 2024, entitled “Method for Reducing Nuisance Tripping of Voltage Monitors for DC/DC Converters in Discontinuous Conduction Mode (DCM),” which is hereby incorporated by reference.

BACKGROUND

[0002] A switch mode converter is an electronic circuit that converts an input direct current (DC) voltage into one or more DC output voltages that are higher or lower in magnitude than the input DC voltage. A switch mode converter that generates an output voltage lower than the input voltage is termed a buck or step-down converter. A switch mode converter that generates an output voltage higher than the input voltage is termed a boost or step-up converter.

[0003] Some switch mode converter topologies include a drive/power switch coupled at a switch terminal to an energy storage inductor/transformer. Electrical energy is transferred through the energy storage inductor/transformer to a load by alternately opening and closing the switch as a function of a switching signal. The amount of electrical energy transferred to the load is a function of the ON/OFF duty cycle of the switch and the frequency of the switching signal. Switch mode converters are widely used in electronic devices, particularly battery powered devices, such as portable cellular phones, laptop computers, and other electronic systems in which efficient use of power is desirable.

SUMMARY

[0004] In one example, a circuit includes an input terminal, a reference voltage circuit, a comparator, a delay circuit, a conduction mode control circuit, and a switching converter controller. The input terminal is configured to receive a converter output voltage. The reference voltage circuit has an output. The comparator has a first input coupled to the input terminal, a second input coupled to the output of the reference voltage circuit, and an overvoltage output. The delay circuit has an input coupled to the overvoltage output, and a delayed overvoltage output. The conduction mode control circuit has a first input coupled to the overvoltage output, a second input coupled to the delayed overvoltage output, and a mode output. The switching converter controller has a continuous conduction mode (CCM) input coupled to the mode output.

[0005] In another example, a circuit includes a switching converter controller, a comparator, a delay circuit, and a conduction mode control circuit. The switching converter controller is configured to control switching of a switch mode converter circuit. The comparator is configured to compare a switch mode converter output voltage to an overvoltage threshold voltage, and provide an overvoltage signal indicating that the switch mode converter output voltage is greater than the overvoltage threshold voltage. The delay circuit is coupled to the comparator. The delay circuit is configured to delay the overvoltage signal, and provide a delayed overvoltage signal. The conduction mode control circuit has inputs coupled to the comparator and the delay circuit, and a mode output coupled to an input the switching converter controller. The conduction mode control circuit is configured to provide a CCM control signal based on the overvoltage signal and the delayed overvoltage signal. The switching converter controller is configured to transition from operation in a discontinuous conduction mode to operation in a continuous conduction mode responsive to the CCM control signal.

[0006] In a further example, a system includes a power source, a switch mode converter circuit,

and an advanced driver assistance system. The power source has an output. The switch mode converter circuit has an input coupled to the output of the power source. The advanced driver assistance system has an input coupled to the output of the switch mode converter circuit. The switch mode converter circuit includes a reference voltage circuit, a comparator, a delay circuit, a conduction mode control circuit, and a switching converter controller. The reference voltage circuit has an output. The comparator has a first input coupled to the output of the switch mode converter circuit, a second input coupled to the output of the reference voltage circuit, and an overvoltage output. The delay circuit has an input coupled to the overvoltage output, and a delayed overvoltage output. The conduction mode control circuit has a first input coupled to the overvoltage output, a second input coupled to the delayed overvoltage output, and a mode output. The switching converter controller has a continuous conduction mode (CCM) input coupled to the mode output.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a block diagram of an example switch mode converter that includes conduction mode control based on the converter output voltage.

[0008] FIG. 2 is a block diagram of an example conduction mode control circuit suitable for use in the switch mode converter of FIG. 1.

[0009] FIGS. 3 and 4 are graphs of example signals in the conduction mode control circuit of FIG. 2.

[0010] FIG. 5 is a block diagram of an example system that includes the switch mode converter of FIG. 1.

DETAILED DESCRIPTION

[0011] FIG. 1 is a block diagram of an example switch mode converter **100** that includes conduction mode control based on the converter output voltage. The switch mode converter **100** is a buck converter in the example of FIG. 1, and includes a high-side switching transistor **102**, a low-side switching transistor **104**, an inductor **106**, a voltage divider **108**, an output capacitor **110**, a switching converter controller **112**, and a mode control circuit **114**. The high-side switching transistor **102** is coupled between voltage input terminal and a switching terminal, and the low-side switching transistor **104** is coupled between the switching terminal and a reference terminal (e.g., ground). The switch mode converter **100** powers a load **111**.

[0012] The switching converter controller **112** controls the high-side switching transistor **102** and the low-side switching transistor **104** to charge and discharge the inductor **106**. When the high-side switching transistor **102** is turned on and the low-side switching transistor **104** is turned off, current flows from through the high-side switching transistor **102** to charge the inductor **106**. When the low-side switching transistor **104** is turned on and the high-side switching transistor **102** is turned off, current flows through the low-side switching transistor **104** as the inductor **106** is discharged. Current flowing from the inductor **106** charges the output capacitor **110** and powers the load **111**.

[0013] The voltage divider **108** is coupled to an output terminal **100A** of the switch mode converter **100**. The voltage divider **108** divides the output voltage (V_{out}) of the switch mode converter **100** to generate a feedback voltage (VFB) that is proportional to V_{out} . The switching converter controller **112** compares VFB to a reference voltage to generate an error signal used to control switching of the high-side switching transistor **102** and low-side switching transistor **104**, a produce a desired value of output voltage V_{out} .

[0014] The high-side switching transistor **102** and the low-side switching transistor **104** form the power stage of the switch mode converter **100**. The switching converter controller **112** can control switching of the high-side switching transistor **102** and the low-side switching transistor **104** to provide operation in a continuous conduction mode (CCM) or a discontinuous conduction mode

(DCM). In CCM, the current flowing through the inductor **106** is continuous, with current either increasing or decreasing in a switching cycle. In DCM, the power stage may be in a high-impedance state with both the high-side switching transistor **102** and the low-side switching transistor **104** conducting no current. The switching converter controller **112** may provide DCM operation if the current drawn by the load **111** is very low. The reduced switching of DCM operation can increase the efficiency of the switch mode converter **100** with small load currents. However, when operating in DCM, the switch mode converter **100** may source current to the inductor **106**, but may not sink current from the inductor **106**. With no ability to sink current, if the current drawn by the load **111** drops suddenly while operating in DCM, Vout may rise. For example, Vout may increase to a voltage that is greater than the safe operating voltage of the load **111**. Some systems may include circuitry that monitors Vout and turns off the switch mode converter **100** or disconnects the switch mode converter **100** from the load **111** if Vout exceeds a safe operating voltage of the load **111**, which can cause the load **111** turn off or malfunction. [0015] The mode control circuit **114** is coupled between the output terminal **100A** and the switching converter controller **112** to detect an increase in Vout that may be due to reduction in load current while operating in DCM. If such an increase in Vout is detected, the mode control circuit **114** can cause the switching converter controller **112** to revert to operation in CCM. When operating in CCM, the switch mode converter **100** can source current to the inductor **106**, and can sink current from the inductor **106**. Accordingly, if Vout increases due to a reduction in load current while the switch mode converter **100** is operating in DCM, the mode control circuit **114** can cause the switch mode converter **100** to reduce Vout by reverting to CCM operation for a short time. After Vout has been reduced to less than a threshold voltage (e.g., less than a selected overvoltage threshold), the mode control circuit **114** may allow the switch mode converter **100** to return to DCM operation.

[0016] The mode control circuit **114** has an input coupled to the output terminal **100A** (via the voltage divider **108**) for receiving a representation of Vout. The mode control circuit **114** has an output coupled to a conduction mode input of the switching converter controller **112**. The mode control circuit **114** provides a CCM control signal (CCM CONTROL) at the output of the mode control circuit **114** to cause the switching converter controller **112** to switch from DCM to CCM operation if Vout exceeds an overvoltage threshold.

[0017] While the switch mode converter **100** is illustrated as a buck converter, implementations of the mode control circuit **114** may be applied in a boost converter, a buck-boost converter, or other type of single or multi-phase switch mode converter. Implementations of the switch mode converter **100** may be used in a wide variety of applications that benefit from accurate current sensing. For example, the switch mode converter **100** may be used in various applications that may benefit from or require compliance with functional safety standards.

[0018] FIG. 2 is a block diagram of an example mode control circuit **114** suitable for use in the switch mode converter **100**. The mode control circuit **114** includes a reference voltage circuit **202**, a comparator **204**, a delay circuit **206**, and conduction mode control circuit **208**. The reference voltage circuit **202** generates a reference voltage VREF that defines an overvoltage threshold.

[0019] The comparator **204** has a first input coupled to the output of the reference voltage circuit **202** for receiving the overvoltage threshold voltage, and a second input coupled to the output terminal **100A** via the voltage divider **108** for receiving VFB. The comparator **204** compares VFB (which is representative of Vout) provided by the voltage divider **108** to the overvoltage threshold provided by the reference voltage circuit **202**. An output signal OV, provided at the output of the comparator **204**, indicates whether Vout is greater than the overvoltage threshold. The comparator **204** provides OV in a first state (e.g., logic high, or overvoltage state) indicating that Vout is greater than the overvoltage threshold, and provides OV in a second state (e.g., logic low) indicating that Vout is not greater than the overvoltage threshold.

[0020] An input of the delay circuit **206** is coupled to the output of the comparator **204**. The delay

circuit **206** has an output at which a delayed overvoltage signal (DOV) is provided. The delay circuit **206** may provide the DOV in the first state (e.g., a logic high, or overvoltage state) if the OV has the first state for a selected delay time. For example, if the OV maintains the first state for the selected delay time (e.g., a selected transient suppression time), then the delay circuit **206** may cause DOV to transition from the second state (e.g., logic low) to the first state (e.g., logic high). In one example, the delay circuit **206** can be implemented as a timer that is incremented if the OV has the first state and the DOV has the second state, and reset if the OV is in the second state in some examples. Other implementations of the delay circuit **206** may be implemented with different circuitry (e.g., charging a capacitor to a threshold) in other examples. The delay time provided by the delay circuit **206** may be greater than 100 microseconds in some examples.

[0021] The conduction mode control circuit **208** has a first input coupled to the output of the comparator **204** for receipt of OV, and second input coupled to the output of the delay circuit **206** for receipt of DOV. The conduction mode control circuit **208** generates the CCM CONTROL signal based on OV and DOV. An output of the conduction mode control circuit **208** is coupled to the input of the switching converter controller **112**. The conduction mode control circuit **208** may generate CCM CONTROL as shown in Table 1.

TABLE-US-00001 TABLE 1 OV DOV CCM CONTROL 0 0 0 0 1 0 1 0 1 1 1 0

[0022] Accordingly, the conduction mode control circuit **208** sets CCM CONTROL to the first state (e.g., logic one, or CCM state) if OV has the first state (indicating that Vout exceeds the overvoltage threshold) and DOV has the second state (indicating that the delay time started with assertion of OV has not expired). The conduction mode control circuit **208** sets CCM CONTROL to the second state (e.g., logic zero, or DCM state) in all other state combination of OV and DOV. The switching converter controller **112** operates in CMM responsive to CCM CONTROL having the first state. For example, if the switching converter controller **112** is operating in DCM, and the switching converter controller **112** provides CCM CONTROL in the first state, the switching converter controller **112** can transition from operation in DCM to operation in CCM to attempt to pull-down (reduce) Vout. If the conduction mode control circuit **208** provides CCM CONTROL in the second state, the switching converter controller **112** may operate in DCM if appropriate based on load current.

[0023] FIGS. 3 and 4 are graphs of example signals in the switching converter controller **112**. The signals VREF, VFB, OV, DOV, and CCM CONTROL as shown in FIGS. 3 and 4. In FIG. 3, VFB is a scaled representation of Vout. Vout and VFB are increasing at time **302**, and VFB exceeds VREF, indicating that Vout exceeds the overvoltage threshold. The switch mode converter **100** is operating in DCM at time **302**. At time **304**, the comparator **204** sets OV to a logic high. The timing of transition of OV from logic low to logic high may be delayed (as shown in FIG. 3) due to hysteresis applied in the comparator **204**. At time **304**, the conduction mode control circuit **208** sets CCM CONTROL to a logic high based on the states of OV and DOV (as shown in Table 1). In the interval **306**, between transition of OV to logic high and transition of DOV to logic high, the switching converter controller **112** is causing the switch mode converter **100** to operate in CCM, which allows the switch mode converter **100** to draw current from the output terminal **100A**. In FIG. 3, CCM operation does not reduce Vout, and VFB does not fall below VREF. For example, in FIG. 3 the increase in Vout was not caused by a reduction current drawn by the load **111**. At time **308**, the delay circuit **206** sets DOV to a logic high, and the conduction mode control circuit **208** sets CCM CONTROL to a logic low in response. With CCM CONTROL set to a logic low, the switching converter controller **112** may revert to DCM.

[0024] In FIG. 4, VFB is a scaled representation of Vout. Vout and VFB are increasing at time **402**, and VFB exceeds VREF, indicating that Vout exceeds the overvoltage threshold. The switch mode converter **100** is operating in DCM at time **402**. At time **404**, the comparator **204** sets OV to a logic high. The timing of transition of OV from logic low to logic high may be delayed (as shown in FIG. 4) due to hysteresis applied in the comparator **204**. At time **404**, the conduction mode control

circuit **208** sets CCM CONTROL to a logic high based on the states of OV and DOV (as shown in Table 1). In the interval **406**, while CCM CONTROL has a logic high state, the switching converter controller **112** is causing the switch mode converter **100** to operate in CCM, which allows the switch mode converter **100** to draw current from the output terminal **100A**. In FIG. **4**, CCM operation reduces Vout, and at time **408** VFB falls below VREF. At time **410**, the comparator **204** sets OV to a logic low state responsive to the reduction of Vout and VFB. The timing of transition of OV from logic high to logic low may be delayed (as shown in FIG. **4**) due to hysteresis applied in the comparator **204**. At time **410**, the conduction mode control circuit **208** reverts CCM CONTROL to logic low state, and in response, the switching converter controller **112** May revert to DCM operation. At time **412**, the selected delay time expires (relative to time **404**), and DOV remains in a logic low state.

[0025] FIG. **5** is a block diagram of an example system **500** that includes the switch mode converter **100**. The system **500** also includes a power source **502** and an advanced driver assistance system (ADAS) **504**. The power source **502** may be a battery, such as a battery that powers vehicle electrical/electronic systems. The power source **502** provides voltage Vin at an output of the power source **502**. The output of the power source **502** is coupled to an input of the switch mode converter **100**. The switch mode converter **100** generates Vout from Vin. The output of the switch mode converter **100** (e.g., output terminal **100A**) is coupled to an input of the ADAS **504**. The ADAS **504** can apply Vout to power electronic circuits (e.g., processors, memory, sensors, display, etc.) of the ADAS **504**. Other implementations of the system **500** may include different circuitry in place of the ADAS **504** (e.g., an infotainment system, an instrument cluster system, an industrial control system, etc.)

[0026] Because the switch mode converter **100** includes the mode control circuit **114**, the switch mode converter **100** can reduce the incidence of disruption in Vout provided to the ADAS **504** caused reductions in current drawn by the ADAS **504** while the switch mode converter **100** is operating in DCM. Accordingly, the operation of the system **500** may be more reliable than operation of similar systems that lack the switch mode converter **100**.

[0027] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0028] As used herein, the terms “terminal,” “node,” “interconnection,” “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0029] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0030] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) (n-type transistor) or a p-channel

FET (PFET)) (p-type transistor)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0031] References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input (or transistor control terminal) is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

[0032] References herein to a FET being “ON” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “OFF” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

[0033] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same terminals. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two terminals as the single resistor or capacitor.

[0034] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0035] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0036] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: an input terminal configured to receive a converter output voltage; a reference voltage circuit having an output; a comparator having a first input coupled to the input terminal, a second input coupled to the output of the reference voltage circuit, and an overvoltage output; a delay circuit having an input coupled to the overvoltage output, and a delayed overvoltage output; a conduction mode control circuit having a first input coupled to the overvoltage output, a second input coupled to the delayed overvoltage output, and a mode output; and a switching

converter controller having a conduction mode input coupled to the mode output.

2. The circuit of claim 1, wherein: the conduction mode control circuit is configured to provide a CCM control signal at the mode output; and the conduction mode control circuit is configured to set the CCM control signal to a first state responsive to an overvoltage signal at the overvoltage output having the first state; and the switching converter controller is configured to operate in a continuous conduction mode responsive to the CCM control signal in the first state.

3. The circuit of claim 2, wherein: the conduction mode control circuit is configured to change the CCM control signal from the first state to a second state responsive to the overvoltage signal changing from the first state to the second state; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the second state.

4. The circuit of claim 2, wherein: the conduction mode control circuit is configured to change the CCM control signal from the first state to a second state responsive to the overvoltage signal having the first state, and a delayed overvoltage signal at the delayed overvoltage output having the first state; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the second state.

5. The circuit of claim 4, wherein: the comparator is configured to compare the converter output voltage to an overvoltage threshold voltage; and the overvoltage signal having the first state indicates that the converter output voltage is greater than the overvoltage threshold voltage.

6. The circuit of claim 5, wherein the delay circuit is configured to set the delayed overvoltage signal to an overvoltage state responsive to the overvoltage signal indicating that the converter output voltage is greater than the overvoltage threshold voltage for a selected transient suppression time.

7. The circuit of claim 2, further comprising a transistor having a first terminal coupled to a switch mode converter output, a second terminal coupled to a reference terminal, and a control terminal coupled to an output of the switching converter controller; wherein the switching converter controller is configured to cause the transistor to conduct current from the switch mode converter output responsive to the CCM control signal having the first state.

8. A circuit comprising: a switching converter controller configured to control switching of a switch mode converter circuit; a comparator configured to compare a switch mode converter output voltage to an overvoltage threshold voltage, and provide an overvoltage signal indicating that the switch mode converter output voltage is greater than the overvoltage threshold voltage; a delay circuit coupled to the comparator, the delay circuit configured to delay the overvoltage signal, and provide a delayed overvoltage signal; and a conduction mode control circuit having inputs coupled to the comparator and the delay circuit, and a mode output coupled to the switching converter controller, the conduction mode control circuit configured to cause the switching converter controller to transition from operation in a discontinuous conduction mode to a continuous conduction mode responsive to the overvoltage signal and the delayed overvoltage signal.

9. The circuit of claim 8, wherein: the conduction mode control circuit is configured to provide a CCM control signal at the mode output; and the conduction mode control circuit is configured to set the CCM control signal to a CCM state responsive to the overvoltage signal indicating that the switch mode converter output voltage is greater than the overvoltage threshold voltage; and the switching converter controller is configured to operate in a continuous conduction mode responsive to the CCM control signal in the CCM state.

10. The circuit of claim 9, wherein: the conduction mode control circuit is configured to change the CCM control signal from the CCM state to a DCM state responsive to the overvoltage signal indicating that the switch mode converter output voltage is not greater than the overvoltage threshold voltage; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the DCM state.

11. The circuit of claim 9, wherein the conduction mode control circuit is configured to change the

CCM control signal from the CCM state to a DCM state responsive to the overvoltage signal indicating that the switch mode converter output voltage is greater than the overvoltage threshold voltage, and the delayed overvoltage signal indicating that an overvoltage is present after a delay time; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the DCM state.

12. The circuit of claim 8, wherein the delay circuit is configured to set the delayed overvoltage signal to an overvoltage state responsive to the overvoltage signal indicating that the switch mode converter output voltage is greater than the overvoltage threshold voltage for a selected transient suppression time.

13. The circuit of claim 9, further comprising a power stage having an output coupled to an input of the comparator and an input coupled an output of the switching converter controller, the switching converter controller configured to cause the power stage to sink current through the output responsive to the CCM control signal having the CCM state.

14. A system comprising: a power source having an output; a switch mode converter circuit having an input coupled to the output of the power source; and an advanced driver assistance system having an input coupled to the output of the switch mode converter circuit, the switch mode converter circuit including: a reference voltage circuit having an output; a comparator having a first input coupled to the output of the switch mode converter circuit, a second input coupled to the output of the reference voltage circuit, and an overvoltage output; a delay circuit having an input coupled to the overvoltage output, and a delayed overvoltage output; a conduction mode control circuit having a first input coupled to the overvoltage output, a second input coupled to the delayed overvoltage output, and a mode output; and a switching converter controller having a continuous conduction mode (CCM) input coupled to the mode output.

15. The system of claim 14, wherein the conduction mode control circuit is configured to cause the switching converter controller to operate in the CCM responsive to an overvoltage signal provided at the overvoltage output and a delayed overvoltage signal provided at the delayed overvoltage output.

16. The system of claim 14, wherein: the conduction mode control circuit is configured to provide a CCM control signal at the mode output; and the conduction mode control circuit is configured to set the CCM control signal to a first state responsive to an overvoltage signal at the overvoltage output having the first state, and a delayed overvoltage signal at the delayed overvoltage output have a second state; and the switching converter controller is configured to operate in the CCM responsive to the CCM control signal in the first state.

17. The system of claim 16, wherein: the conduction mode control circuit is configured to change the CCM control signal from the first state to a second state responsive to the overvoltage signal changing from the first state to the second state; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the second state.

18. The system of claim 16, wherein: the conduction mode control circuit is configured to change the CCM control signal from the first state to a second state responsive to the overvoltage signal having the first state, and the delayed overvoltage signal having the first state; and the switching converter controller is configured to operate in a discontinuous conduction mode responsive to the CCM control signal in the second state.

19. The system of claim 16, wherein: the comparator is configured to compare a representation of a converter output voltage provided at the output of the switch mode converter circuit to an overvoltage threshold voltage; and the overvoltage signal having the first state indicates that the representation of a converter output voltage is greater than the overvoltage threshold voltage.

20. The system of claim 16, wherein the delay circuit is configured to set the delayed overvoltage signal to indicate an overvoltage responsive to the overvoltage signal having the first state for a selected transient suppression time.

